Plotter enclosures · cut & bend pack

Two stacked electronics boxes (control on top, power below), each a folded body + a folded lid, plus two flat base plates and three flat acrylic window pieces. Prepared by Allwin Tom · 2026-10-14 · v5

#	File	Part	Material	Qty	Flat (mm)	Bends	Folded outside (mm)
1	`control_body.dxf`	CONTROL BOX · BODY	1.0 mm 304 stainless, brushed, film on	1	699.0 × 434.8	11	452 W × 297 D × 111 H
2	`control_lid.dxf`	CONTROL BOX · LID	1.0 mm 304 stainless, brushed, film on	1	484.5 × 423.5	6	456 W × 300.0 D × 111 H (back)
3	`control_base_plate.dxf`	CONTROL BOX · BASE PLATE	2.0 mm aluminium 5052 (or 1.5 mm 304)	1	440.0 × 285.0	0	flat, 440 × 285, notched
4	`power_body.dxf`	POWER BOX · BODY	1.0 mm 304 stainless, brushed, film on	1	819.0 × 494.8	11	452 W × 297 D × 171 H
5	`power_lid.dxf`	POWER BOX · LID	1.0 mm 304 stainless, brushed, film on	1	484.5 × 483.5	6	456 W × 300.0 D × 171 H (back)
6	`power_base_plate.dxf`	POWER BOX · BASE PLATE	2.0 mm aluminium 5052 (or 1.5 mm 304)	1	440.0 × 285.0	0	flat, 440 × 285, notched
7	`acrylic_power_left.dxf`	POWER FRONT · WINDOW ACRYLIC, LEFT	3 mm clear cast acrylic, protective film on both sides	1	149.0 × 56.0	0	flat
8	`acrylic_power_right.dxf`	POWER FRONT · WINDOW ACRYLIC, RIGHT	3 mm clear cast acrylic, protective film on both sides	1	149.0 × 56.0	0	flat
9	`acrylic_control.dxf`	CONTROL FRONT · WINDOW ACRYLIC	3 mm clear cast acrylic, protective film on both sides	1	149.0 × 56.0	0	flat

Bending

- Every flat is drawn from the **inside face**. Every bend is **UP 90°**, inner radius **R1**.
- Flats assume K 0.44, bend allowance 2.262 mm, bend deduction 1.738 mm per 90° bend (1 mm 304, R = t). **If your bend deduction for 1 mm 304 at R1 differs, tell us the number**: we re-issue the flats so every hole lands where it should.
- Bend order: small lips first (top lips, back lips, corner tabs, base back lip), then the side walls, then the front wall last. Lid: return lips first, then the top-to-back bend.
- Corner reliefs and end-of-flange reliefs are already cut into the outline.
- **Deliberate gaps, please keep:** flange ends stop 2.5 mm short of neighbouring flanges; side-wall front edges and corner tabs stop 0.5 mm short of the front wall (the rivets close it); the lid is 1.0 mm per side larger than the body.

Layers

- **Visible** — cut (outline, holes, cutouts, vents).
- **Bend Center** (dash-dot) and **Visible Narrow** (bend-zone edges) — bend references, **do not cut**.
- **Bend Notes**, **Leaders**, **Title** — information only, do not cut or engrave.
- DXF R12, millimetres, 1:1.

Finish & handling

- 304 stainless, brushed, protective film left on. Grain the same direction on all visible faces of both boxes.
- Deburr all edges. No paint or coating.
- Base plates: 2 mm 5052 aluminium (1.5 mm 304 acceptable). Deburr.
- Acrylic: 3 mm clear cast acrylic, laser cut, film left on both sides. Edges are hidden, no polishing.
- Tolerances: laser ±0.2 mm, bends ±0.5 mm.

Fitted by the customer — do not insert

- M3 rivnuts (Ø5 holes) in the body lips, round the control-front connector openings, round all three windows, round the IEC and USB side-panel cutouts, on the control lid top and the power front; solid rivets Ø3 (Ø3.2 holes) in the front corner tabs.
- Window bezels, connector plates and side panels are 3D printed and screwed on later: into the rivnuts, except the receiving and PSU panels (Ø3.4 holes, screw + nut). The keyhole slot in the middle of the power front takes a printed logo mount.
- M4 rivnuts (Ø6 grid) in the base plates as needed; M4 standoffs, rubber feet, panels, fans.

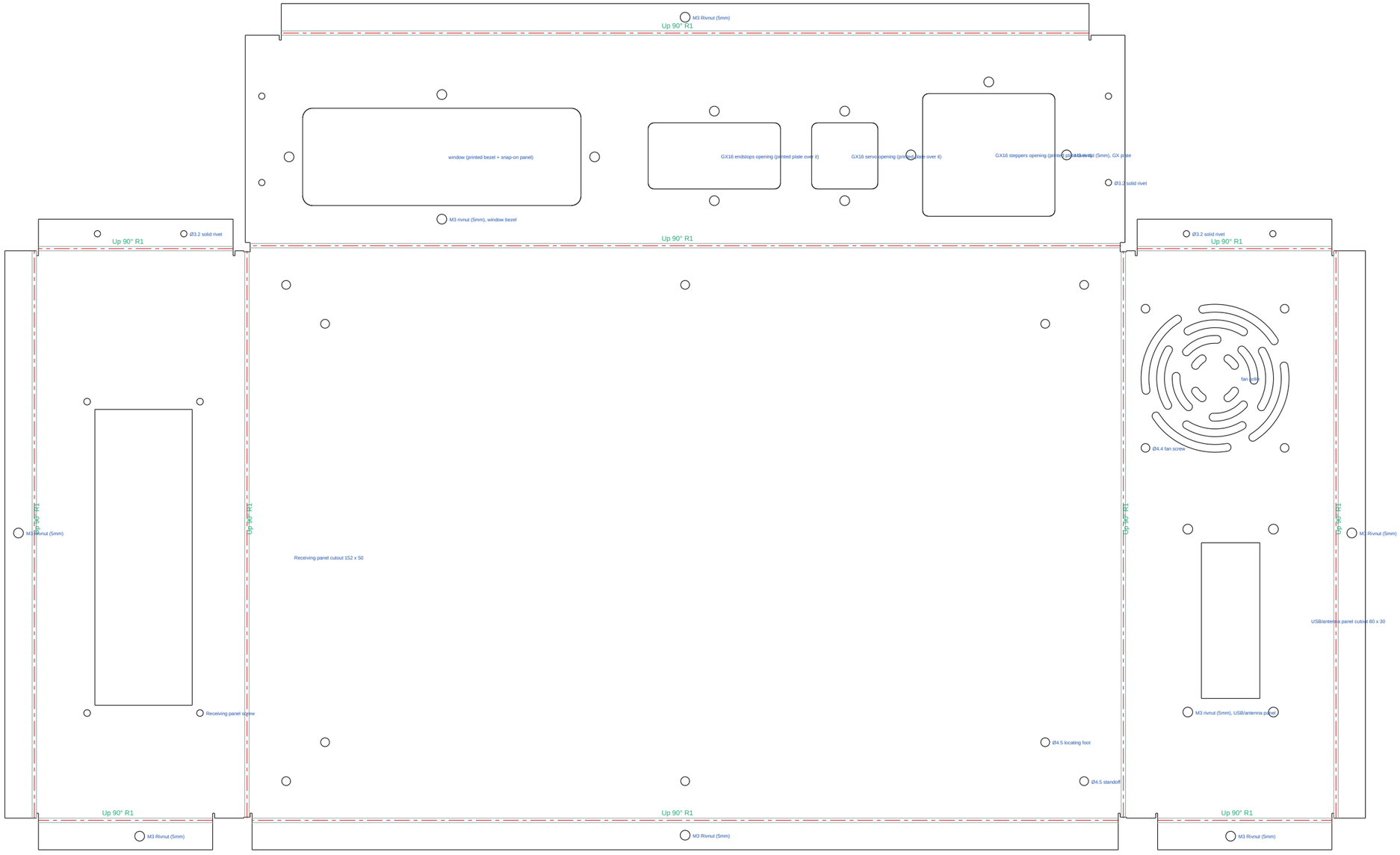
Checks: 806 design checks passed before these files were written — holes clear of bends (≥ 5 mm), webs ≥ 1.5 mm, folded parts free of clashes, folded flanges at least 2 mm apart wherever they are not meant to touch, whole assembly (fans, panels, rivnuts, base plate, standoffs) clear by 2 mm, lid clear of the body, every lid hole over its rivnut, base-plate standoffs over the floor's, panel cutouts and screw patterns matched to the printed panels, every bend end relieved. The files were then read back and folded in software from the DXFs alone (bends UP, R1): every hole lands where the design puts it, no part touches another, and with an inner radius anywhere from R0.6 to R2.5 the lid still fits. Re-fold from their own bend lines:

- refold body width: 452.00 (design 452.00)
- refold left wall height: 111.00 (design 111.00)

- refold right wall height: 111.00 (design 111.00)
- refold body depth (base): 297.00 (design 297.00)
- refold front wall height: 111.00 (design 111.00)
- refold lid width: 456.00 (design 456.00)
- refold lid depth: 300.00 (design 300.00)
- refold body width: 452.00 (design 452.00)
- refold left wall height: 171.00 (design 171.00)
- refold right wall height: 171.00 (design 171.00)
- refold body depth (base): 297.00 (design 297.00)
- refold front wall height: 171.00 (design 171.00)
- refold lid width: 456.00 (design 456.00)
- refold lid depth: 300.00 (design 300.00)

1 · CONTROL BOX · BODY

`control_body.dxf` · 1.0 mm 304 stainless, brushed, film on · qty 1 · flat 699.0 × 434.8 mm · folded 452 W × 297 D × 111 H



CONTROL BOX - BODY

Material: 1.0 mm 304 stainless, brushed, film on Qty: 1 Units: mm

Drawn from the INSIDE face. All bends: UP 90°, inner radius R1.

Flat assumes K 0.44, bend allowance 2.26, deduction 1.74 per bend. Shop's value wins: re-run if different.

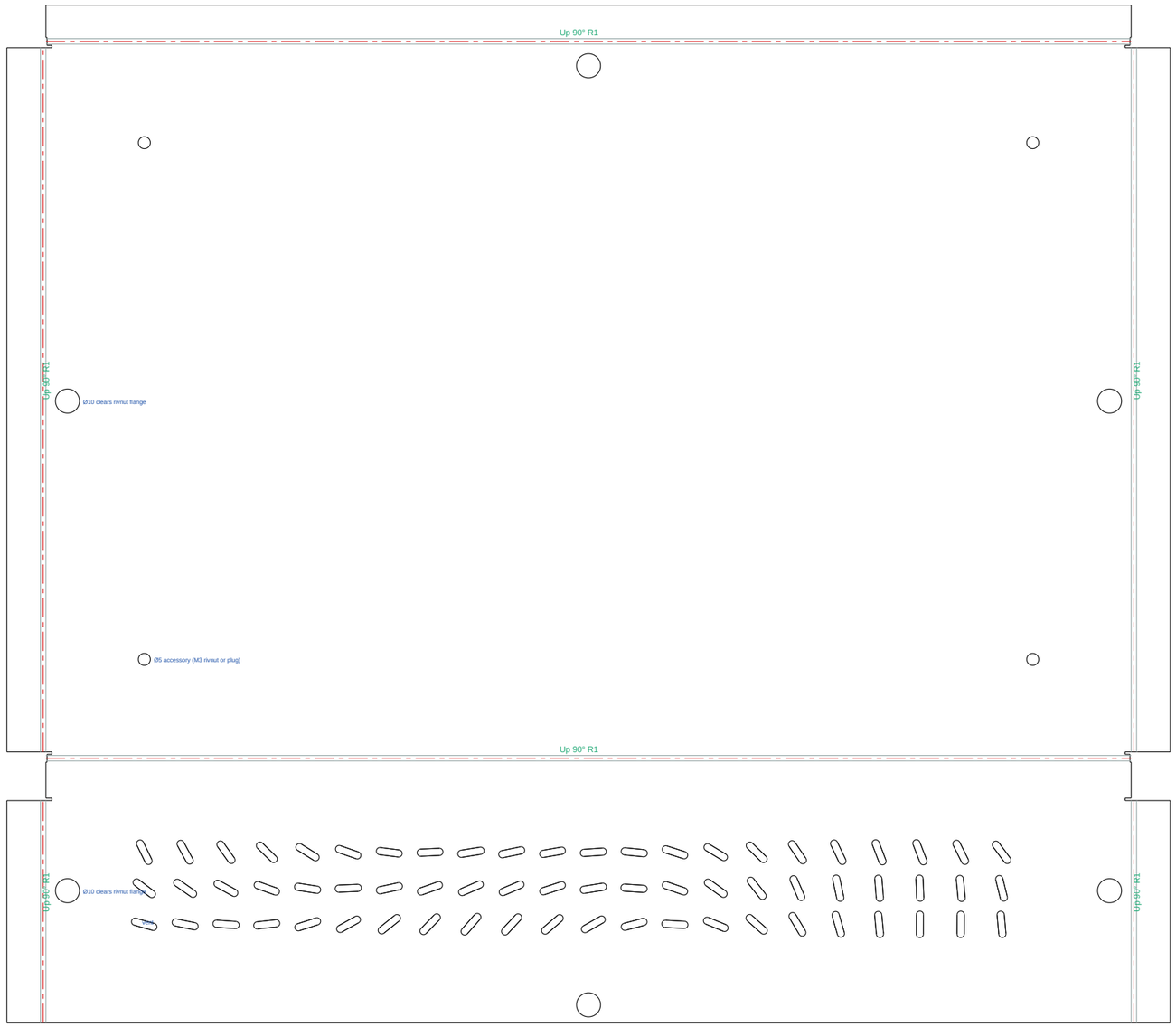
Folded outside: 452 W x 297 D x 111 H (body, lips in)

Inside: 450 x 295 x 110

Flat size: 699.0 x 434.8

2 · CONTROL BOX · LID

`control_lid.dxf` · 1.0 mm 304 stainless, brushed, film on · qty 1 · flat 484.5 × 423.5 mm · folded 456 W × 300.0 D × 111 H (back)



CONTROL BOX - LID

Material: 1.0 mm 304 stainless, brushed, film on Qty: 1 Units: mm

Drawn from the INSIDE face. All bends: UP 90°, inner radius R1.

Flat assumes K 0.44, bend allowance 2.26, deduction 1.74 per bend. Shop's value wins: re-run if different.

Folded outside: 456 W x 300.0 D x 111 H at the back

Fits over the body with 1.0 mm per side inside the return lips

Flat size: 484.5 x 423.5

3 · CONTROL BOX · BASE PLATE

`control_base_plate.dxf` · 2.0 mm aluminium 5052 (or 1.5 mm 304) · qty 1 · flat 440.0 × 285.0 mm · folded flat, 440 × 285, notched



CONTROL BOX - BASE PLATE

Material: 2.0 mm aluminium 5052 (or 1.5 mm 304) Qty: 1 Units: mm

Flat part, no bends.

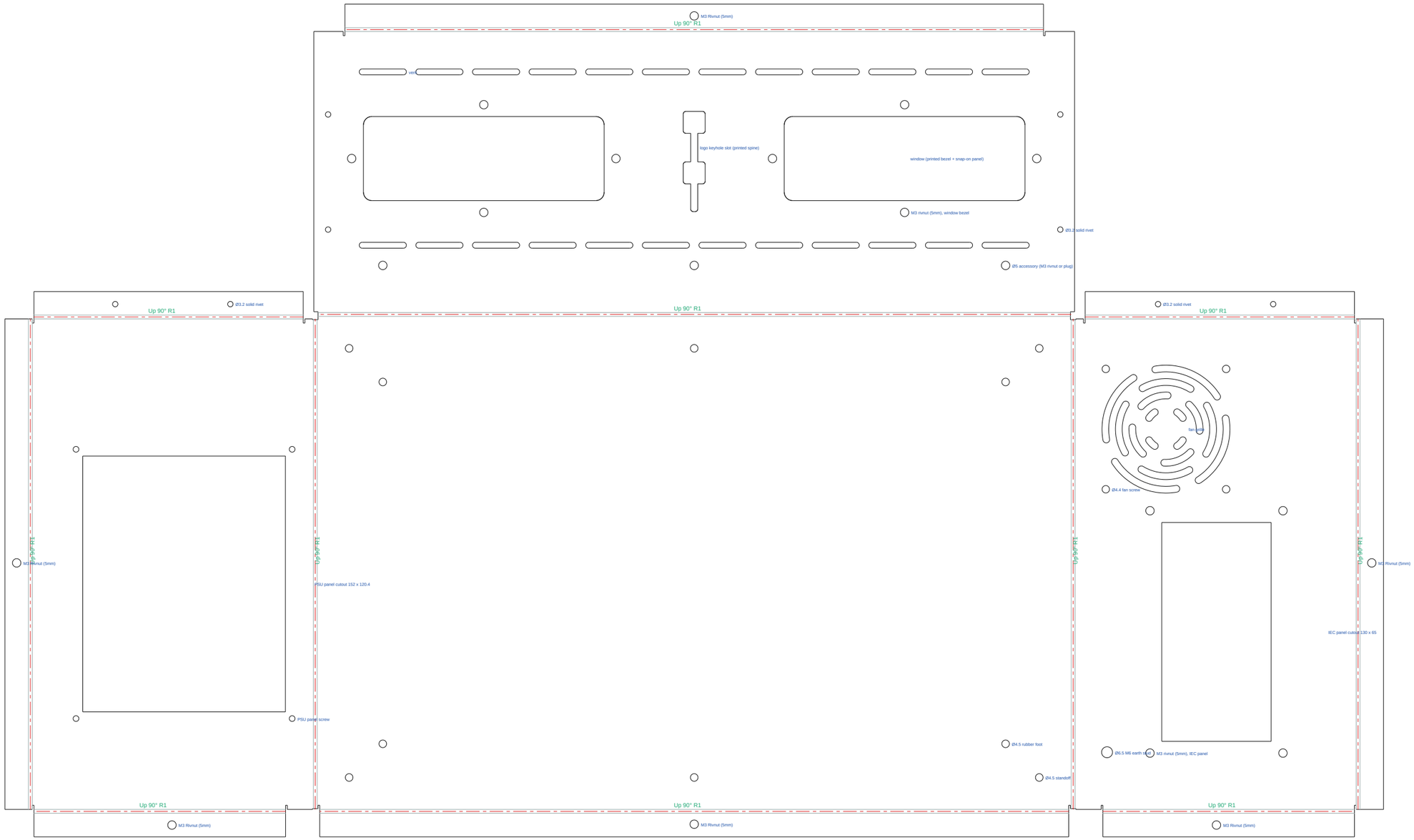
Flat, no bends. Sits on 6 x M4 standoffs, 5 mm.

Grid: 40 mm rows, every other row shifted 20 mm

Flat size: 440.0 x 285.0

4 · POWER BOX · BODY

`power_body.dxf` · 1.0 mm 304 stainless, brushed, film on · qty 1 · flat 819.0 × 494.8 mm · folded 452 W × 297 D × 171 H



POWER BOX - BODY

Material: 1.0 mm 304 stainless, brushed, film on Qty: 1 Units: mm

Drawn from the INSIDE face. All bends: UP 90°, inner radius R1.

Flat assumes K 0.44, bend allowance 2.26, deduction 1.74 per bend. Shop's value wins: re-run if different.

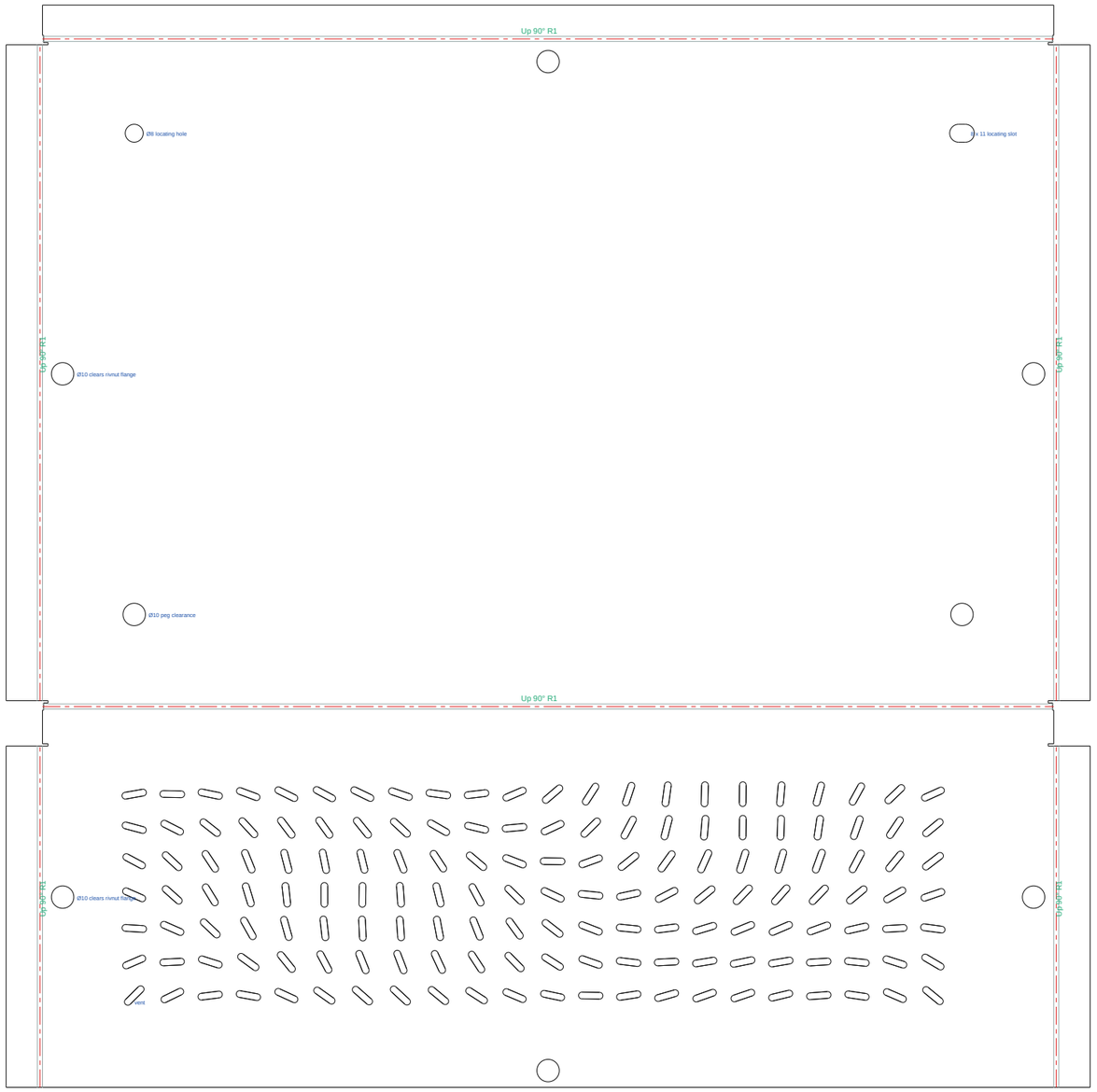
Folded outside: 452 W x 297 D x 171 H (body, lips in)

Inside: 450 x 295 x 170

Flat size: 819.0 x 494.8

5 · POWER BOX · LID

`power_lid.dxf` · 1.0 mm 304 stainless, brushed, film on · qty 1 · flat 484.5 × 483.5 mm · folded 456 W × 300.0 D × 171 H (back)



POWER BOX - LID

Material: 1.0 mm 304 stainless, brushed, film on Qty: 1 Units: mm

Drawn from the INSIDE face. All bends: UP 90°, inner radius R1.

Flat assumes K 0.44, bend allowance 2.26, deduction 1.74 per bend. Shop's value wins: re-run if different.

Folded outside: 456 W x 300.0 D x 171 H at the back

Fits over the body with 1.0 mm per side inside the return lips

Flat size: 484.5 x 483.5

6 · POWER BOX · BASE PLATE

`power_base_plate.dxf` · 2.0 mm aluminium 5052 (or 1.5 mm 304) · qty 1 · flat 440.0 × 285.0 mm · folded flat, 440 × 285, notched



POWER BOX - BASE PLATE

Material: 2.0 mm aluminium 5052 (or 1.5 mm 304) Qty: 1 Units: mm

Flat part, no bends.

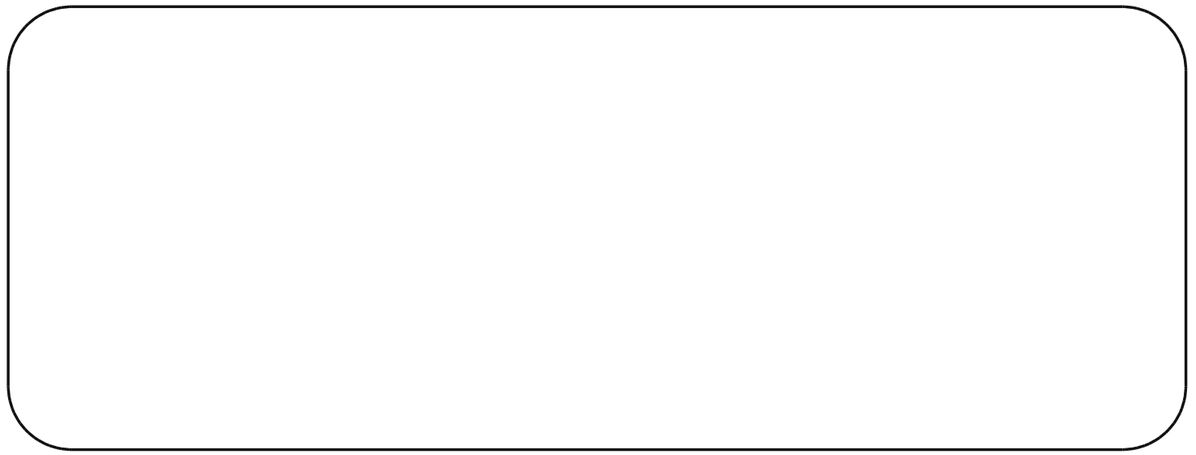
Flat, no bends. Sits on 6 x M4 standoffs, 5 mm.

Grid: 40 mm rows, every other row shifted 20 mm

Flat size: 440.0 x 285.0

7 · POWER FRONT · WINDOW ACRYLIC, LEFT

`acrylic_power_left.dxf` · 3 mm clear cast acrylic, protective film on both sides · qty 1 · flat 149.0 × 56.0 mm · folded flat



POWER FRONT - WINDOW ACRYLIC, LEFT

Material: 3 mm clear cast acrylic, protective film on both sides Qty: 1 Units: mm

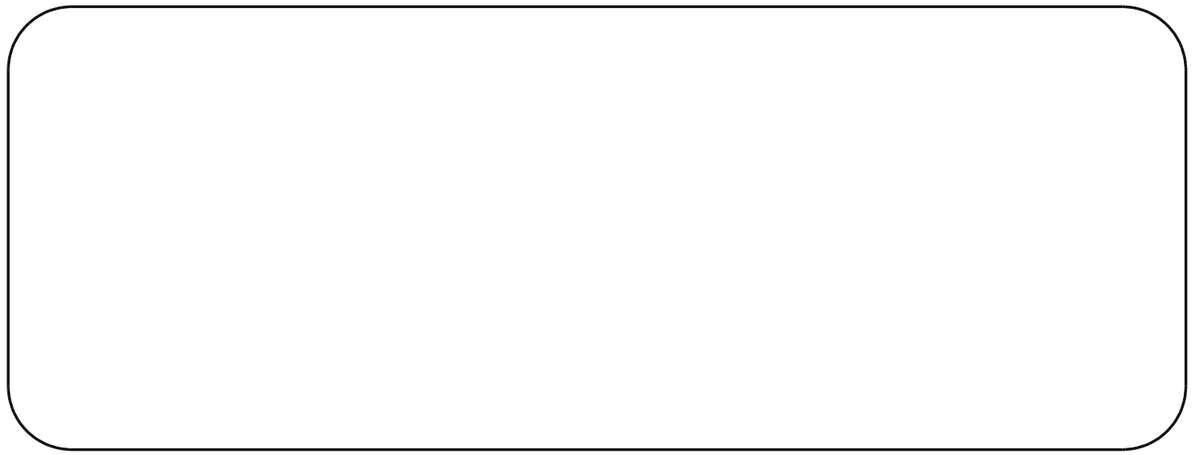
Flat part, no bends.

Laser cut. Edges hidden under a printed bezel: no polishing needed.

Flat size: 149.0 x 56.0

8 · POWER FRONT · WINDOW ACRYLIC, RIGHT

`acrylic_power_right.dxf` · 3 mm clear cast acrylic, protective film on both sides · qty 1 · flat 149.0 × 56.0 mm · folded flat



POWER FRONT - WINDOW ACRYLIC, RIGHT

Material: 3 mm clear cast acrylic, protective film on both sides Qty: 1 Units: mm

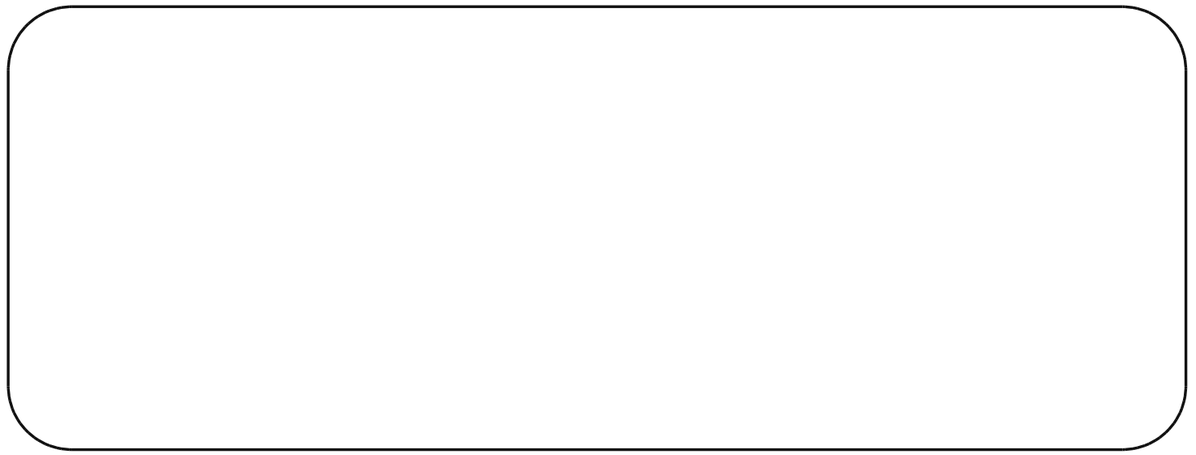
Flat part, no bends.

Laser cut. Edges hidden under a printed bezel: no polishing needed.

Flat size: 149.0 x 56.0

9 · CONTROL FRONT · WINDOW ACRYLIC

`acrylic_control.dxf` · 3 mm clear cast acrylic, protective film on both sides · qty 1 · flat 149.0 × 56.0 mm · folded flat



CONTROL FRONT - WINDOW ACRYLIC

Material: 3 mm clear cast acrylic, protective film on both sides Qty: 1 Units: mm

Flat part, no bends.

Laser cut. Edges hidden under a printed bezel: no polishing needed.

Flat size: 149.0 x 56.0